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Using ChatGPT as a Lesson Planning Assistant with Preservice Secondary Mathematics Teachers

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Abstract

The purpose of this qualitative study was to examine the use of ChatGPT as a lesson-planning assistant with preservice teachers (PSTs) of secondary mathematics. PSTs asked ChatGPT to solve a mathematics problem and to assist with lesson planning in a microteaching context in a methods course. The PSTs first developed parts of their microteaching lesson plan individually, then asked ChatGPT to do the same thing. The PSTs then reflected on the output generated by ChatGPT. They then used ChatGPT any way they saw fit throughout the planning process. An analysis of the PSTs' reflective statements about ChatGPT's output revealed the importance of critical and evaluative reflection. Although PSTs were generally accurate in their assessment of pedagogical output, noting that ChatGPT's suggested lessons were teacher-centered and repetitive, indicating little knowledge of students' needs; they were less accurate about their assessment of mathematical output, often attributing incorrect solutions as a "different approach" to the problem. The findings of this study have implications for the value of a reflective process for PSTs when using ChatGPT and other chatbots for lesson planning and solving mathematics problems, and suggest that a framework for critical examination of mathematical output may be necessary in the age of GAI.

Keywords Generative AI · ChatGPT · Secondary mathematics · Preservice teachers

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Introduction

Generative AI (GAI) chatbots such as ChatGPT are poised to fundamentally change the work of teachers, and therefore teacher preparation programs are at the forefront of integrating the use of GAI in educational settings (MacDowell & Korchinsk, 2023). GAI is poised to change how teachers develop assessments, plan lessons, create word problems, and influence other components of teaching. For example, ChatGPT may be used to grade student work and create progress reports for students. It can also translate information to facilitate communication with non-English-speaking students and parents (Trust et al., 2023), to determine the readability of text, to track and monitor student progress, and to provide feedback (Celik et al., 2022). Thus, it is essential that preservice teacher education programs incorporate the use of GAI chatbots, so that preservice teachers (PSTs) are prepared to use these technologies in productive and useful ways as they enter the teaching profession (Tate et al., 2023). Although GAI can be useful, it is imperfect, especially in mathematics (Wardat et al., 2023), and teachers, particularly PSTs, must be reflective about the output generated by chatbots such as ChatGPT. Tate et al. assert that GAI is "fallible but useful" (2023, p. 7).

One use of GAI is to generate lesson plans and to make suggestions regarding pedagogical approaches (Trust et al., 2023). Another use of GAI is to prompt it to solve mathematics problems, so that PSTs are aware of different possible solutions to the problems, although GAI is sometimes incorrect, performing at the level of an average graduate student (Frieder et al., 2024). Thus, using ChatGPT's output requires reflective analysis on the part of the user. Lesson planning is of particular interest, since researchers have found that PSTs have difficulty planning lessons (Cevikbas et al., 2024; Lim et al., 2018; Sahin-Taskin, 2017). Many PSTs first encounter the requirement to plan a lesson in a methods class, often through a microteaching experience. This study examines the incorporation of ChatGPT, specifically version GPT-3.5, into a lesson-planning experience in a methods course, where PSTs were instructed to use ChatGPT as an assistant for writing a lesson plan for a microteaching experience referred to as a "mini-lesson". In this study, PSTs interacted with ChatGPT to obtain ideas for lesson planning, and to examine solutions to the mathematics problem used to launch their mini-lesson. Therefore, the purpose of this study is to answer the following research questions: (1) How did PSTs react to the mathematical output produced by ChatGPT? (2) What did PSTs find useful/not useful when using ChatGPT to plan a mini-lesson?

First, we discuss theoretical frames for teacher reflection and attitudes toward the use of GAI, then the literature on PSTs' issues with lesson planning, and the literature on microteaching, then discuss the use of GAI such as ChatGPT for mathematics, and for lesson planning. We then discuss the procedures of the study and data analysis. Finally, we present results and discuss implications for the field regarding the use of GAI in teacher preparation.

Literature Review and Theoretical Framework

Theoretical Framework

The theoretical frame for this study draws upon the work of Artzt (1999), Knoth et al. (2024), and Celik et al. (2022). We used the Artzt framework because it is infused throughout the authors' teaching program, is familiar to the PSTs, and holds reflective practice at the center of every aspect of instruction. This frame asserts that instructional practice (which mainly concerns tasks as in the present study, but also, learning environment, and discourse) is a byproduct of teachers' knowledge, beliefs, and goals. These components drive teachers' preactive thoughts (lesson planning), interactive thoughts (monitoring of lesson), and postactive thoughts (reflective ideas about improving the lesson).

The framework emphasizes the critical importance of teachers' knowledge and beliefs about mathematics, pedagogy, and students. Further, the Artzt framework emphasizes the importance of student, rather than teacher-centered instruction. Although this may not be internationally accepted, student-centered instruction is emphasized as a goal in the USA (and in local districts) given the findings of the TIMSS studies and recommendations of the NCTM (NCTM, 2000; Gurl et al., 2016). The work of Knoth et al. is a framework for AI literacy, both generic and domain-specific, the domain in the present study being education.

Similar to the Artzt frame, Knoth et al. include cognition (knowledge), behavior (skills), and attitudes (values) about AI and its capabilities and limitations. Celik et al. found in their review of AI-related literature that AI-related advantages for teachers are related to planning, implementation, and assessment of lessons, while disadvantages are related to lack of reliability of AI output, limited access, and lack of technical knowledge. Although the Celik et al. study was not focused on *generative* AI, the findings are applicable to the use of GAI.

The main emphasis for this study is planning for components of the mathematical task, and examining the PSTs' reflections related to the use of ChatGPT for planning. Components of tasks are modes of representation, motivation, difficulty level, and sequencing (Artzt, 1999; Artzt & Armour-Thomas, 1999; Artzt et al., 2015). Modes of representation are "forms for representing mathematical concepts", while motivation refers to tasks that "attract and sustain (learners') attention and emotional investment" (Artzt et al., 2015, p. 14). Difficulty level and sequencing refers to the "complexity … and order in which the teacher makes the task salient for (students)" (Artzt et al., 2015, p. 15).

The main emphasis of this study is on the tasks, although other components of the framework also emerged at times as part of the PSTs' reflective process. The work of Knoth et al. parallels that of Artzt since both emphasize the role of cognition/knowledge, along with beliefs/attitudes. Knoth et al., however, emphasize the need for AI literacy, an essential facet of this study. Some of the findings of this study extend and in some cases mildly contradict those of Knoth et al. The findings of Celik et al. are framed around planning, implementation, and assessment, also parallel to the Artzt framework. These three studies provide a robust frame for examining PST interactions with ChatGPT.

Literature on PST Lesson Planning and Microteaching

Lesson planning, which is "the thinking that teachers do as they prepare to teach lessons" (Artzt et al., 2015, p. 25), is often quite difficult for PSTs (Cevikbas et al., 2024; Lilly et al., 2024; Lim et al., 2018; Sahin-Taskin, 2017). Planning a lesson should result in a document referred to a "lesson plan" which includes an outline of the important components of the lesson. In our program, this includes an aim, objectives, motivational questions, tasks and activities, and assessment. PSTs' difficulties in lesson planning are often related to a lack of knowledge about students' misconceptions and how students think about mathematical concepts and designing or applying real-life applications of what they were teaching into their lessons (Cevikbas et al., 2024). Even high-performing PSTs were much more skillful in enacting lessons but "have yet to develop skills in lesson planning" (Lim et al., 2018, p. 528). Further, PSTs often find it difficult to create their own plans and find suitable activities and problems for students, although there is some evidence for the use of technology to facilitate lesson planning (Cevikbas et al., 2024). The initial steps for planning are especially difficult for PSTs, particularly finding suitable activities, and these difficulties often persist for early-career teachers as well (Lilly et al., 2024; Sahin-Taskin, 2017).

Given PSTs' difficulties with lesson planning, they often benefit from a scaffolded approach to learning how to plan a lesson. Lim et al. (2018) found that PSTs struggle with creating their own lessons, and performed better when modifying existing lesson plans or synthesizing components of lessons, rather than creating their own lessons from scratch. Lilly et al. (2024) found that early-career teachers improved their lesson planning when they were able to make explicit connections between the mathematics content they were teaching and knowledge of how students would react.

Another scaffold that many teacher preparation programs often incorporate is microteaching. Microteaching is a form of practice lesson planning and teaching to peers that often takes place in a methods course (Bell, 2007) and "provide(s) PSTs with more authentic opportunities to enhance their knowledge and skillset" (Jakopovic et al., 2023, p. 29). Microteaching is often complementary to traditional field experiences and may contribute to a teacher's improved sense of self-efficacy. Furthermore, microteaching includes opportunities for peer feedback and PST reflection (Arsal, 2014; Jakopovic et al., 2023).

The literature specific to the planning process as related to microteaching is less conclusive. Although microteaching always involves planning a lesson, the process by which this occurs is not a prevalent in research findings or discussion. While Bell (2007) simply mentions that the microteaching process allows PSTs to have some experience planning a lesson, Arsal (2014) and Jakopovic et al. (2023) say a bit more, including the benefit of PSTs setting gaining experience identifying learning objectives and standards for their plans. According

to Arsal (2014), PSTs became more efficient in planning and developed self-confidence in planning, while the PSTs in Jakopovic et al. (2023)'s study found the opportunity to practice writing lesson plans beneficial. Therefore, it is clear that any additional support for PSTs in writing lesson plans, and opportunities to practice writing plans, would be beneficial.

Literature on GAI for Lesson Planning

There is a growing literature base related to using GAI such as ChatGPT in educational contexts. ChatGPT is a large language model which is trained on large datasets and is able to generate text (Marr, 2023). It can assist teachers in creating lesson plans, assessments, translating language for non-native speakers, creating instructional materials and can even function as a mentor for novice teachers (Lo, 2023). Celik et al. (2022) found that AI can provide information about students' backgrounds and make suggestions about appropriate content and activities. ChatGPT, however, is known to have varying levels of performance, particularly in mathematics (Frieder et al., 2024; Shryock, 2023), and may be getting worse in its ability to do mathematics correctly (Chen et al., 2023). However, there is also the chance it will improve. In the present study, the uneven quality of the GAI output provides ample opportunity for the PSTs to reflect critically.

An important drawback to using ChatGPT in planning mathematics lessons is that it often produces incorrect mathematical output (Wardat et al., 2023) and has been found to be limited in its ability to plan entire lessons (Gurl et al., in process). There have, however, been recent findings that there were better student outcomes when teachers used lessons produced by ChatGPT rather than lessons prepared by teachers alone (Karaman & Goksu, 2024). Kehoe (2023) found that the benefits of using ChatGPT for lesson planning included saving time, increasing creativity, and personalization of lessons but emphasized the need for quality control and awareness of the inherent bias in GAI algorithms. Thus, there remains much uncertainty about the usefulness of using ChatGPT to plan lessons and do mathematics at this time.

What is certain, however, is that PSTs need to be able to critically reflect on the correctness, usefulness, and overall value of output produced by GAI chatbots. Although GAI output can be incorrect, especially when doing mathematics, it is likely that PSTs (and others) will continue to use it, since, as Tate et al. (2023) state, "the proverbial cat is already out of the bag", and even though there are weaknesses, there are also "powerful affordances" (p. 8). As stated in our theoretical frame, reflection is at the center of all aspects of teaching (Artzt, 1999; Artzt & Armour-Thomas, 1999; Artzt et al., 2015), and PSTs' cognition, attitudes, and behavior (Knoth et al., 2024) contribute to their critical reflection about and evaluation of ChatGPT's output in the context of their teaching.

Procedures

Subjects and Context

Subjects of this qualitative study were a convenience sample of PSTs ($n = 22$) who were enrolled in either of two sections of a secondary mathematics methods course at a public, urban, university in the northeastern US. Seventeen of the PSTs were undergraduate students, and five were graduate career changers. A method course requirement was for each PST to plan and teach a mini-lesson (i.e., microteaching) to the rest of the class, who role-played as secondary students, and to the professors, who are the second and third authors of this work. The PSTs were required to select their mini-lesson problem from problem-solving tasks from the class textbooks (Gurl et al., 2012, 2013) and then develop a lesson using the task as a motivation for the lesson. Although each PST did an individual mini-lesson, there were two sections of the course, so some topics were planned and taught by two different individuals. The lessons were taught twice, concurrently, but in two different classrooms, because there were two different sections of the course taught by two different professors (i.e., the second and third author of this work).

The professors and authors of this work emphasized a problem-solving approach to teaching as consistent with the standards of the National Council of Teachers of Mathematics (NCTM) in the US. The NCTM's standards recommend using problems with students to "build new mathematical knowledge" (NCTM, 2000, p. 52). The preface of the required texts note that the tasks are recommended to be used as launch problems for teaching (Gurl et al., 2012, 2013).

Data Collection and Analysis

In order to support the PSTs in planning their mini-lessons over the course of several weeks, data were collected in two phases with professor (and ChatGPT) support. Once each PST chose their mathematical task, the first phase included a homework assignment at the early planning stages, well before teaching the mini-lesson. The assignment was for each PST to complete their chosen mini-lesson mathematical task with no assistance, and to begin the lesson planning process by writing a focus question (referred to as an "aim" as consistent with requirements of local districts) and behavioral objectives for their lesson.

As part of the same homework assignment, they were then asked to engage with ChatGPT by doing the following and entering responses to these Phase 1 items into an online form: *(1) Ask ChatGPT to do your mini-lesson task. How do your solutions compare? (2) Ask ChatGPT to formulate an Aim and objectives for a secondary mathematics lesson based on your mini-lesson task. Copy and paste the output. Feel free to modify your prompts as you see fit, but be sure to attach the entire output. (3) Re-write your Aim and objectives for your mini-lesson after using ChatGPT as an assistant. If you are not making any changes to your Aim or objectives, explain why. (4) Reflection: How did using ChatGPT affect your Aim and Objectives? Did*

you find consulting with ChatGPT to be similar to or different from working with a peer or a professor? Explain.

This initial interaction with ChatGPT took place during September 2023. Over the next few weeks while developing their lesson plans, PSTs were instructed to continue to use ChatGPT as a resource for the planning process in any way they saw fit, but there was no requirement that they use it any particular number of times. Over the semester, each PST taught their lesson to their class and professor, submitting pre- and post-lesson thoughts, lesson plans, and activities. Two or three mini-lessons took place per class meeting, and the class met once per week. After they taught, the PSTs responded to the following Phase 2 items as part of their post-lesson thoughts: *Summarize your impression of your interactions with ChatGPT regarding your mini-lesson. State at least one example of what ChatGPT provided that was and was not useful. Please include screenshots of your exchange with ChatGPT and explain your reasoning.* The phases of data collection are summarized in Table 1.

Data from this study were analyzed using qualitative coding methods as described by Charmaz (2006) and Saldaña (2016). According to Charmaz, "coding means naming segments of data with a label that simultaneously categorizes, summarizes and accounts for each piece of data" (2006, p. 43). Data were coded incident to incident; that is, each PST's response to a data collection prompt was considered to be a single "incident" that was assigned a deductive code that was determined prior to analysis, although we also looked for new ideas that might emerge from the data. Saldaña recommends deductive coding when the "inquiry is theory-driven and targets specific experiences, phenomena, actions, and so on that you are certain will appear in the empirical materials" (2016, p. 39). For example, when coding the mathematical solutions, we knew we were looking for

Table 1 Phases of data collection

Phase	PST task
1	PSTs selected and solved their problem individually, then asked ChatGPT to solve the same problem, and compared the solutions. PSTs then wrote the beginning of their lesson (aim and objectives) alone, then with the assistance of ChatGPT. PSTs continued using ChatGPT informally to plan their lessons, including the option of modifying their original aim and objectives PSTs then reflected on its use
2	After teaching, PSTs summarized impressions of the usefulness of ChatGPT after continuing to use it independently to plan the lesson. [Post-lesson thoughts]

correctness; when examining the pedagogical output, we expected to see evidence of Artzt's framework since that was heavily emphasized in the methods class.

Data for the first research question related to the mathematical output (e.g., content) were drawn from the PSTs' and ChatGPT's solutions for the mathematics task, along with the PSTs' comparison of the solutions in Phase 1, as described above. The first author used incident to incident coding to determine correctness for each PST's solution, followed by determining correctness of ChatGPT's associated mathematical output and then each PST's reflection about ChatGPT's output for accuracy of evaluation. A solution was coded as "correct" if it was completely correct, or if the only errors were minor, such as arithmetic, and there were no conceptual errors. The first author then examined ChatGPT's solution to the same problem, and coded ChatGPT's solution for correctness using the same standard. The first author then coded each PST's written response that asked them to compare their solution to that of ChatGPT.

All three authors then met to discuss and defend the coding results to ensure reliability. Results for the first research question (regarding mathematical content) are first organized by whether only the PST, only ChatGPT, or both obtained a correct answer, then how the PST compared their answer to that of ChatGPT. In summary, there were 19 instances of a PST doing a problem and then prompting ChatGPT to do the same problem. Then, the PST's solution and ChatGPT's solution were each coded for correctness, and the PST's reaction to ChatGPT's solution was coded for whether they identified the correctness of ChatGPT's output, and how they compared ChatGPT's output. Although there were 22 PSTs in the study, three of them did not correctly follow instructions; therefore, those PST responses could not be included in the analysis.

In order to answer the second research question regarding what the PSTs found useful and not useful when planning, the first author deductively coded their responses to the Phase 1 and Phase 2 reflective questions, first coding responses to the Phase 1 question: *How did using ChatGPT affect your Aim and Objectives? Did you find consulting with ChatGPT to be similar to or different from working with a peer or a professor? Explain.* This was coded incident by incent, by determining whether the PST stated that they made any changes to their initial attempts at lesson planning, then coding whether they thought working with ChatGPT was different or similar to working with a professor or peer.

The first author then coded responses to the Phase 2 reflective prompt: *Summarize your impression of your interactions with ChatGPT regarding your mini-lesson. State at least one example of what ChatGPT provided that was and was not useful. Please include screenshots of your exchange with ChatGPT and explain your reasoning.* The responses to this item were deductively coded for what was stated by the PSTs to be useful and not useful as either being related to content or pedagogy, and then what aspects of the theoretical framework were evident in terms of planning for tasks, learning environment, or discourse (Artzt et al., 2015). The second and third authors then met with the first author, coming to a consensus on the coding as related to the deductive codes.

Table 2 Summary of correct and incorrect mathematical responses to the same task

PST response	ChatGPT response	Number
Correct	Correct	5
Correct	Incorrect	10
Incorrect	Incorrect	3
Incorrect	Correct	1
Not coded		3
Total		22

Table 3 Middle school task 1.4

Percent: Kara was shopping at her favorite store which is holding a sale that offers 20% off the price of every item in the store. She had a coupon that gives her an additional 50% off the sale price. She thinks this is great, because now she will get 70% off of the items she buys

a) Is Kara correct? Explain
b) If Kara wants to buy an item that originally costs $60, how much will the item cost given both discounts?
c) If the tax rate is 7.5%, what will be the final cost of the item in part (b)?
d) Which is the better deal: 20% off with an additional 50% off the sale price, or 50% off with an additional 20% off the sale price? (Gurl et al., 2013, p. 8)

Results

This section will be organized by the results for each of the research questions, first reporting the PSTs' reactions to the mathematical output and then by discussing what the PSTs found useful and not useful when using ChatGPT to plan a mini-lesson. For each, a discussion of the coded results will be followed by specific statements by the PSTs and output generated by ChatGPT to better illustrate the PSTs' interactions with the chatbot as viewed through the lens of the reflective framework (Artzt, 1999; Artzt & Armour-Thomas, 1999; Artzt et al., 2015).

PSTs' Reactions to ChatGPT's Mathematical Output

There were 19 PST and associated ChatGPT responses coded for the mathematical problems, which included algebra, geometry, data analysis, and percent, at the middle- and high-school level (US grades 7–12). Of these, 15 of the PSTs' mathematical solutions were coded as "correct", while only six of the ChatGPT responses were coded as "correct". We then compared the code for the PST solution to the code for ChatGPT solution for the same problem. For the 19 PSTs' responses, coding revealed five instances where the PST and ChatGPT solutions were both correct, ten where the PST was correct, but ChatGPT's response was not, three where both were incorrect, and one where the PST was incorrect and ChatGPT was correct. These results are summarized in Table 2.

We then coded the PST reflections about the ChatGPT mathematical output. Among the five instances where both the PST and ChatGPT correctly solved the problem, the PST reactions said just that, mentioning that ChatGPT's solutions were the same, but some noted that different mathematical approaches were used, along with different language and formatting. An example is the PST's reflection about task 1.4. The task is reproduced in Table 3, followed by the PST's response.

The PST solved the problem correctly, and stated the following in their reflection about ChatGPT's correct solution to the same problem:

> The solution to part b and c of my task were the same. But when it comes to part a and d, it took a different approach. For part a, instead of showing the difference in prices using the two different methods, it multiplied the discount and shows what the actual combined percent she gets off the original item. And, for part d, I state the commutative property of multiplication for the order of application not mattering, but the ai (sic) showed that using each method the same amount would be taken off the original item. (PST response)

This was an accurate assessment of the differences between responses, where both the PST and ChatGPT were correct. The PST applied their mathematical knowledge while reflecting on the output, noting differences in the approach to the problem. As stated above there were only five instances where both the PST and ChatGPT solved the problem correctly.

There were 10 instances where the PST solved the problem correctly while ChatGPT did not. In all but one of these 10 instances, the PSTs did not correctly identify the issues with ChatGPT's response. First, we will share the one instance where the PST correctly evaluated and reflected upon ChatGPT's incorrect output. We will then share three instances where the PST did not correctly identify the errors in ChatGPT output, two for the same task. For each instance, the task will be shared, then ChatGPT's output with the incorrect solution, along with the PST's reflection about and evaluation of the generated solution.

The PST who correctly evaluated ChatGPT's incorrect output completed task 1.4, shown in Table 4.

This was a problem that examined the relationship between the perimeter and area of a rectangle represented by a polynomial, which the PST answered correctly. The PST then prompted ChatGPT to solve the problem. The single prompt was the statement, "The following is an activity for a high school level algebra lesson. Answer the questions in the activity and explain the solutions", followed by the exact statement of the task as in Table 4. Table 5 contains ChatGPT's output for this task, with incorrect output underlined, as added by the researchers.

Table 4 High school task 1.4

The perimeter of a rectangle is represented by $4x+24$. In response to the question "What is a possible representation of the area of this rectangle?" Mark says x^2+36, and Sarah says no, $x^2+12x+36$
What assumptions are both Mark and Sarah making about the rectangle? Comment on their answers
Other students are also discussing the problem. Alex says the area is $24x$. Anna says that she thinks the area is $x^2+12x+27$. Where do you think they are getting these answers from? Can they all be correct?

Table 5 ChatGPT solution to high school task 1.4

ChatGPT

Let's break down the activity and analyze the responses from Mark, Sarah, Alex, and Anna:

1. **Mark's Response:** He suggests that the area of the rectangle is represented by x^2 + 36
2. **Sarah's Response:** She disagrees with Mark and suggests that the area is x^2 + 12x + 36

Now, let's consider what assumptions Mark and Sarah are making about the rectangle:

- **Mark's Assumption:** Mark assumes that the rectangle is a square (because he doesn't account for the 12×term in the area expression). In a square, all sides are equal, so the area formula would indeed be x^2 where x is the length of each side
- **Sarah's Assumption:** Sarah assumes that the rectangle is not necessarily a square but rather a rectangle with a length of x + 6 and a width of x + 6 (since x + 6 and x + 6 combined equals 4x + 24, which is the perimeter). Therefore, she includes both the length and width when calculating the area, leading to x^2 + 12x + 36

Now, let's consider the responses from Alex and Anna:

3. **Alex's Response:** Alex suggests that the area is 24x. This answer seems to assume that the rectangle has no width (x) and consists of a single line with length24x. This is not a correct representation of a rectangle, as it lacks width
4. **Anna's Response:** Anna suggests that the area is x^2 + 12x + 27. Similar to Sarah, Anna assumes that the rectangle is not necessarily a square and uses an expression that includes both the length and width

In summary:

- Mark and Sarah have different assumptions about the rectangle's dimensions, which lead to their different area formulas. Mark assumes a square, while Sarah assumes a rectangular shape with a specific length and width
- Alex's answer does not correctly represent a rectangle; it appears to assume that the rectangle has no width, which is not a valid assumption
- Anna's answer is similar to Sarah's and assumes a rectangle with specific dimensions

All of these answers cannot be correct simultaneously because they make different assumptions about the rectangle's shape and dimensions. The correct answer depends on the specific characteristics of the rectangle, and the area formula should be based on those characteristics. If the rectangle is indeed a square, Mark's answer would be correct. If it's a rectangular shape as assumed by Sarah and Anna, their expressions would be appropriate. Alex's answer, however, is not a valid representation of a rectangle's area

The PST who correctly evaluated and reflected upon ChatGPT's incorrect response (and was the only PST to do so) wrote the following:

> The solutions given by ChatGPT are for the most part, entirely wrong. Even when correct, the explanations given make no sense and the AI completely misinterpreted the basic instructions of the activity. I believe the lack of units (since my activity is about the area and perimeter of rectangles) caused even more confusion for the AI because it incorrectly states that 24x can not be a valid representation of a rectangle'ds area, even though obviously it can be the area of a rectangle with adjacent side lengths of 24 units and x units respectively. (PST response)

This response was an accurate evaluation of the incorrect response, and the PST made a reflective conjecture about what might have prevented an accurate response, namely that the lack of stated units might have caused confusion for the chatbot, although there is no way to know whether that is true. ChatGPT accurately completed some aspects of the problem but misinterpreted a good deal of what the problem was asking, including essential conditions of the problem. The

Table 6 High school task 1.3 – algebra

Ms. A's class has decided to buy her a gift. They found the perfect item at the store, and it costs G dollars. The students in the class agree to split the cost equally. The next day, 10 students decide that they cannot contribute. Write an algebraic expression for the additional amount the other students had to contribute to buy the same gift
a) What quantities do we know? How might they be represented? Which need to be represented by a variable?
b) Decide which quantities need to be represented by an algebraic expression, and write the expression
c) Write an expression for the difference in price. Make sure it is in simplest form. (Gurl et al., 2012, p. 7)

Table 7 ChatGPT excerpt from solution to high school task 1.3

ChatGPT
Certainly! Let's break down this problem step by step:
[…]
To find the additional amount the other students had to contribute to buy the same gift after 10 students couldn't contribute, we first need to calculate how much each student would have to contribute if everyone participated. Then, we can subtract this from the total cost to find the additional amount
The cost per student if everyone participated is the total cost divided by the total number of students:
Cost per student = G / N
Now, we can calculate the total cost for the students who could participate (N—10 students):
Total cost for participating students = (N—10) * (Cost per student)
To find the additional amount they had to contribute, we subtract this total cost for participating students from the original cost of the gift:
Additional Amount = G—[(N—10) * (G / N)]
To simplify further, we can factor out G/N from the second term:
Additional Amount = G—G * (N—10) / N
Now, let's factor out G as a common term:
Additional Amount = G * [1—(N—10) / N]
Simplify the expression inside the square brackets:
Additional Amount = G * [1—(N/N—10/N)]
Additional Amount = G * [1—(1—10/N)]
Further simplify the expression inside the square brackets:
Additional Amount = G * [1—1 + 10/N]
Now, simplify the expression within the square brackets:
Additional Amount = G * (10/N)
So, the algebraic expression for the additional amount the other students had to contribute to buy the same gift is:
Additional Amount = 10G/N
This is the simplest form of the expression

PST was accurate in their reflection, applying their own mathematical knowledge to the evaluation of ChatGPT's output.

Next, we will share another example where a PST completed their problem correctly, high school task 1.3 (see Table 6), along with ChatGPT's7 incorrect response to the same problem, and the PST's reflection about ChatGPT's output.

The PST correctly and accurately completed the problem, representing the unknown quantities and the difference in cost for the students algebraically. ChatGPT's solution is below in Table 7 with incorrect parts underlined. Most of the correct parts of ChatGPT's solution have been omitted.

The PST's reaction follows. Note that "they" refers to ChatGPT.

> One difference that I saw was that they subtracted the total cost for participating students from the original cost of the gift. Instead of doing that, I subtracted the cost per student of the gift who could not participate from the original cost to figure out the additional cost. I believe our solutions did come out differently because of this. (PST response)

Similar to above, ChatGPT missed essential parameters of the problem. While the PST identified the different procedure, they did not identify it as incorrect, even though they completed the problem correctly, revealing possible issues with their beliefs about their own mathematical knowledge and revealing their beliefs about ChatGPT's "ability" to do mathematics.

In some cases, two PSTs taught a lesson based on the same problem (since there were two sections of the methods class). Two PSTs completed a lesson based on high school task 1.6 (see Table 8). In both cases, the PST completed the task correctly, while ChatGPT did not. In both reflections, the PSTs noticed that ChatGPT's approach was different, but not that it was incorrect, similar to the above. Further, ChatGPT's incorrect output was different for each PST.

Note that each PST entered high school task 1.6 exactly as stated in Table 8. Although it is not possible to determine which of these PSTs interacted with ChatGPT first, it is worth examining the differences in the output generated by ChatGPT for each PST. Both PSTs completed the task correctly, and one of them considered some possible student responses. Both of the incorrect solutions generated by ChatGPT follow, along with the respective PST reflections. These PSTs are referred to as PST 1 and PST 2 for clarity. The output generated by PST 1 is shown in Table 9, and the output generated by PST 2 is in Table 10, with incorrect statements underlined. The graph is also incorrect, but that was not possible to underline. (Although it may seem surprising, this graph is exactly what was generated by ChatGPT as provided by the.html link from the PST.) Note that the output in Table 9 has fewer mistakes than that in Table 10

In this example, ChatGPT generated an equation, then listed coordinate pairs with only positive integer values, used mathematically awkward language, and plotted several points incorrectly.

Although there were many issues with this output, the PST's reflection did not address the errors. PST 1's reaction was simply to say, "My solution was a bit more in-depth and mentioned the slope. ChatGPT's solution only mentioned whether a solution was valid or not and how it looked on the graph. Both solutions were very

Table 8 High school task 1.6 – algebra

The sum of two numbers is ten. List several pairs of numbers that satisfy this requirement.
If each pair of numbers is considered as an ordered pair on the coordinate plane and graphed, what do you notice about the set of graphed points? What if you continue plotting points that follow the same pattern, and write down their coordinates? What is the relationship between the coordinates? What if one of the numbers is negative? a fraction? a negative fraction? greater than 10? (Gurl et al., 2012, p. 13)

Table 9 ChatGPT's response to high school task 1.6 generated by PST 1

ChatGPT

The sum of two numbers being 10 means we're looking for pairs of numbers (x, y) that satisfy the equation:

$x + y = 10$

Here are several pairs of numbers that satisfy this requirement:

(0, 10), (1, 9), (2, 8), (3, 7), (4, 6), (5, 5), (6, 4), (7, 3), (8, 2), (9, 1),(10, 0)

Now, let's plot these points on a coordinate plane and observe the pattern:

```
10 |          x
   |
9 |        x
   |
8 |      x
   |
7 |    x
   |
6 |  x
   |
5 | x
  |x
4 |  x
   |      x
3 |     x
   |         x
2 |        x
   |            x
1 |           x
   |_________________
0 1 2 3 4 5 6 7 8 9 10
```

From the plot, you can observe that the points lie on a straight line. The relationship between the coordinates is that they all satisfy the equation $x + y = 10$, which represents a linear relationship

Now, let's discuss different scenarios:

1. **One of the numbers is negative:**

For example, (-2, 12) and (7, -3). The negative number simply means we're moving in the opposite direction on the coordinate plane

2. **A fraction:**

For example, (3.5, 6.5). Fractions are still valid, and they would lie on the same line

3. **A negative fraction:**

For example, (-1.5, 11.5). Negative fractions are also valid and would still lie on the same line

4. **Greater than 10:**

For example, (15, -5). The point would still lie on the same line, but on the side where x is greater than 10

In summary, any pair of numbers (x, y) that satisfy $x + y = 10$ will lie on a straight line when plotted on a coordinate plane, and this pattern continues for fractions, negative numbers, negative fractions, and numbers greater than 10. The line represents a linear relationship between the two variables

similar." Thus, PST 1 did not mention the incorrect graph nor mention that $(7, -3)$ was said to be a solution. Table 10 shows the ChatGPT output for the same task, generated by PST 2. Incorrect parts are underlined.

When analyzing this output, step 4 reveals some issues based on the statement "one number increases by 1 as the other decreases by 1. In other words, they have

Table 10 ChatGPT's second response to high school task 1.6 generated by PST 2

ChatGPT
Certainly, I can help you explore this problem step by step 1. **Pairs of Numbers That Sum to 10:** (0, 10), (1, 9), (2, 8), (3, 7), (4, 6), (5, 5), (6, 4), (7, 3), (8, 2), (9, 1),(10, 0) 2. **Graphing These Pairs:** If you consider each pair as an ordered pair on the coordinate plane and graph them, you will notice that they all lie on a straight line with a slope of -1. This line passes through the point (5, 5), which represents the pair (5, 5) where both numbers are 5, making the line symmetrical 3. **Continuing the Pattern:** If you continue plotting points that follow the same pattern, you'll notice that they all lie on the same line with a slope of -1. You can extend this line in both directions indefinitely 4. **Relationship Between Coordinates:** The relationship between the coordinates of these points is that one number increases by 1 as the other decreases by 1. In other words, they have a difference of 1. This is because the sum of the coordinates must always be 10 5. **One of the Numbers is Negative:** If one of the numbers is negative, you can still find pairs that satisfy the requirement. For example: - (-1, 11) - (-2, 12) These pairs will also lie on the same line with a slope of -1 6. **A Fraction or a Negative Fraction:** If one of the numbers is a fraction or a negative fraction, you can still find pairs that sum to 10, such as: - (3.5, 6.5) - (-2.5, 12.5) 7. **Greater than 10:** If one of the numbers is greater than 10, then the other number must be negative to maintain a sum of 10. For example: - (15, -5) In summary, the set of graphed points forms a straight line with a slope of -1, passing through the point (5, 5). The coordinates of these points have a difference of 1, and this pattern continues indefinitely in both directions along the line. Whether one of the numbers is negative, a fraction, a negative fraction, or greater than 10, you can still find pairs that satisfy the requirement and they will all lie on the same line

a difference of 1." This statement is problematic for several reasons. Although the statement is loosely related to the idea of slope, or the points themselves when represented as integers and listed chronologically, it is incorrect mathematical language. Further "they have a difference of 1" is completely incorrect. Although one might consider this statement as related to a recursive approach and thus not incorrect, it is vague and not an example of strong mathematical output on the part of ChatGPT. The summary at the end of the output also states "The coordinates of these points have a difference of 1, and this pattern continues indefinitely in both directions along the line." This is related to the incorrect statement in step 4. PST 2's comparison of their own solution to that of ChatGPT follows:

> Our solutions are quite similar. They are almost identical. The only thing I would note is that ChatGPT uses more cohesive and mathematical language. It is not language that a high school student who is seeing this concept for the first time would use in class. They also stated the slope as -1 right at the

> beginning when the task doesn't introduce the concept nor does the task require students to define it until later on. ChatGPT also stated that the line graph is symmetrical and extends in both directions indefinitely, which is something I did not mention (PST 2 response).

Thus, even when entering the same prompt, ChatGPT's output was different, and in both cases, several elements were incorrect. In both cases, the PSTs both noted differences in the solution, but did not identify the incorrect output nor the incorrect mathematical language, even suggesting that ChatGPT's solution was better, as evidenced by the comment about "more cohesive and mathematical language". Further, these reactions may reveal beliefs about each PST's own mathematical knowledge, and possibly their beliefs about students' use of mathematical language, and beliefs about ChatGPT's mathematical language.

PSTs' Reactions to the Usefulness of ChatGPT's Output

To determine what the PSTs found useful and not useful when planning, and answer research question 2, we coded responses to the Phase 1 and Phase 2 items related to pedagogy (see above for Phase 1 and 2 items). We first determined whether the PST stated that they made any changes to their initial attempts at lesson planning, then coded whether they thought working with ChatGPT was different or similar to working with a professor or peer. We then coded the responses to the Phase 2 reflective prompt. The responses to this item were coded for what was stated by the PSTs to be useful and not useful as either being related to content or pedagogy, and then what aspect of the theoretical framework were evident in terms of planning for tasks, learning environment, or discourse (Artzt et al., 2015).

A summary of coding of whether a change was made follows. Of the 22 coded responses, 20 indicated that they changed their aim or objectives in some way, while two did not make any changes. One of the two PSTs who did not make a change stated:

> I am not making any changes to my Aim or my objectives. I found that the objectives ChatGPT listed were very similar to the ones I used in my own assignment. ChatGPT also included objectives that weren't explicitly relevant to the task, such as "Analyzing real-world scenarios" relating to rational and irrational numbers. I also found that phrases such as "Use critical thinking skills" were not measurable and should not be included in objectives. (PST response)

A PST who did make a change stated, "ChatGPT gave me a better understanding of my goal in teaching my mini-lesson, and it affected my objective. I was struggling to come up with my objectives for my mini-lesson, however, ChatGPT was able to list them all out for me." Another PST who made a change stated, "The results of ChatGPT helped me improve my aim and objectives. The results of ChatGPT were more comprehensive and thoughtful." Another PST (who was writing a lesson based

on a task related to earning more points versus performing better in basketball) went into more detail:

> ChatGPT did not affect my Aim, but it greatly affected my Objectives. I liked the way that ChatGPT had more specific objectives that pertained to the lesson. The only aspect that I did not like was relating the entire lesson to the basketball percentages, so I changed the objectives to omit that aspect. I was able to alter the objectives in a way that every scenario could apply to the objectives, not just basketball. After asking ChatGPT to generalize the objectives, they became much more clear and could apply to all quantities of ratios rather than just basketball. This was very helpful because the objectives were separated into categories in a similar manner that is expected of us in our lesson plans. So, when looking for a specific category for an objective, it was easy to find. (PST response)

Each of the above PSTs describes an aspect of how they incorporated ChatGPT's output into their objectives and goals for the lesson, but did not use the output as is.

Of the 22 responses that were analyzed for whether working with ChatGPT was similar to or different from working with a peer or professor, 6 were coded as "similar" or "somewhat similar," 13 were coded as "different" and 3 did not address that part of the question so were not coded. One PST whose response was coded as similar reflected that "Consulting with ChatGPT is similar to working with a peer or a professor because they allow me to ask related questions and explain my task. It makes the work effective." Another PST was a bit more cautionary, noting some limitations of ChatGPT while still saying it was similar to working with a peer or professor, saying.

> ChatGPT is similar to working with a peer and/or professor. It is interesting to see the answers it provides and how it works. However, I think ChatGPT users have to be careful trusting ChatGPT 100%. It's only useful with existing knowledge on the topic. (PST response)

Among the respondents who indicated that working with ChatGPT was different than working with a peer or professor noted the lack of "back-and-forth", stating:

> I found this interaction to be different from consulting with a peer or professor. This interaction was different because there was not much conversation back and forth. Typically, when having a conversation, one does not answer with a 500-word essay. The responses are way more in-depth and lengthy than a typical conversation. This does not leave much room for collaboration as a discussion between a peer or professor would. (PST response)

Another mentioned limitations in ChatGPT's ideas and again, the lack of back-and-forth, stating:

> Consulting with ChatGPT was very different from working with a peer or a professor, because ChatGPT does not actually have its own ideas to give you. There are no thoughts being bounced back and forth and improved upon. It feels much more like working with a tool to make it spit out exactly what you

> want, given whatever you already told it. At least for math-related prompts, ChatGPT doesn't seem very useful beyond optimizing the way sentences are written out. (PST response)

Another PST had similar thoughts, noting that ChatGPT does not ask questions, stating:

> Working with ChatGPT is very different from working with peers or a professor. This is because the AI does not brainstorm nor does it ask me any questions. It simply gives answers. Other people may take time to think or ask questions. The answers ChatGPT gives are all very clean, at least grammatically, whereas people don't always speak or write eloquently 100% of the time. I also don't argue with ChatGPT like I do with my peers. It also writes a lot very fast. Other people take a lot longer to process and come up with as much information. (PST response)

Another noted that working with ChatGPT is different, alluding to the visibility of the thinking process, saying:

> Consulting with ChatGPT is different than working with a peer or professor because when working with AI, they just give me the answers. On the other hand, working with peers and professors, we get to see the thought process behind each idea. (PST response)

The PSTs were insightful in their observations of ChatGPT's limitations. Perhaps this PST summed it up best, noting that it can be a good starting point, stating:

> Consulting with ChatGPT was different from working with a professor or peer because I had to use the responses provided by ChatGPT as a starting point to assess my own work and see how ChatGPT did it better, if at all and then I had to see what responses were useful and which were not. (PST response)

Thus, the PSTs were selective about what they incorporated into their own lessons from the ChatGPT output. None used the generated output as is; instead, they reflected upon the output and considered their pedagogical and mathematical goals for the lesson. Their responses also seemed to indicate that their beliefs about their own mathematical knowledge might have influenced their evaluation of ChatGPT's mathematical output.

Of the 22 PSTs who were subjects of the study, 20 responded to the items that asked them to describe what was useful and not useful when interacting with ChatGPT. When discussing what was useful, 13 of their responses were coded as being related to pedagogy, 6 were coded as being related to content, and one response was coded as related to both content and pedagogy. All respondents discussed something both useful and not useful. What follows is a discussion of what PSTs reflected as useful and what was not.

Some of the things PSTs found useful were related to different aspects of pedagogy. For example, one PST who found that ChatGPT's output related to assessment was useful stated that:

> An example where ChatGPT provided an useful example for my lesson plan was its use of assessment. It stated that to assess students, the teacher should evaluate students based on their participation in geoboard activities, and their ability to decompose triangles and calculate areas correctly. Although this is not the best information for ChatGPT to provide, it gives the teacher an idea of what they are supposed to do when assessing their students. (PST response)

As a result of this interaction with ChatGPT, the following assessment questions were written into this PST's lesson plan, indicating a change related to their assessment goals for students:

> Skill Check #1: Find the area of the triangle with vertices, $(-3, 4)$, $(3, -3)$, and $(5, 1)$. Explain.
> Skill Check #2: Triangle ABC with coordinates $A(0, -6)$, $B(3, -5)$, and $C(6, -6)$ is on the coordinate plane below. List out all possible points that can result in the same area as Triangle ABC. Explain why the points you chose can result in the same area.

Another PST found useful output related to content, specifically, common student misconceptions related to the content of the lesson, stating:

> I asked ChatGPT to come up with some misconceptions that I could put in my lesson plan. It came up with some valid ones that may have come up in a typical high school classroom. Prior to this, I had some of my friends with varied math experiences complete the Do Now task for my lesson, and they held some of these misconceptions as they discussed the task. Therefore, I found ChatGPT useful when it comes to generating misconceptions, as they seem to be accurate. ChatGPT was slightly helpful for a few lesson planning components. I found it useful for coming up with misconceptions and clarifying questions. (PST response)

In fact, this PST did incorporate the ChatGPT output into their lesson plan, including a list of possible misconceptions that was based on the ChatGPT output. Another stated that:

> The discussion (with ChatGPT) was beneficial as it provided me with a structured outline, guiding me on what to include in each section of the lesson plan. It also provided some objectives for my lesson, and I rephrased them using appropriate action verbs. Furthermore, it assisted me in identifying the prior knowledge that students should need for my lesson. It was helpful for me since I was unsure about what objectives to include and what prior knowledge to consider in my lesson. The suggestions provided were valuable, and I incorporated them into my lesson plan.

The following comment from a PST gives a comprehensive explanation of the overall usefulness of ChatGPT, noting that it is a good starting point.

> While we didn't use these exact problems, ChatGPT gave a surprisingly decent variety of problems. With more "prompt artistry", much better results can be obtained from ChatGPT. Of course, this is an example of ChatGPT being very useful for giving the general idea of something that a *human can mold into something much better for their own particular use.* I would not count on using ChatGPT for the final version of any written work, but it is certainly good as a starting point for ideas. ChatGPT seems most useful for giving a rough outline, or listing a broad range of ideas to pick and choose from. While the AI isn't anywhere near perfect, it is useful as a tool for searching and for certain technical linguistic things, like grammar and formatting. (PST response, emphasis added)

This PST explicitly stated what others hinted at—that ChatGPT is useful for output only with human input, and also mentioned the importance of prompting to improve output.

PSTs also discussed what they did not find useful. Of the 20 coded responses, 17 were related to pedagogy, 2 were related to content, and 1 was related to both content and pedagogy. One PST, whose response was related to pedagogy, found that ChatGPT did not generate a useful homework assignment, stating:

> I also tried to use ChatGPT for the creation of my homework assignment: however, I could only use a few of the problems because they were essentially repeated but with different variables. The quality of the problems provided by ChatGPT were all around the same level and not as cognitively challenging as I would have liked. (PST response)

Another PST attempted to use ChatGPT to write an entire plan, which was not satisfactory due to being teacher-centered and not "practical" as evidenced by the following remark.

> I asked ChatGPT to write a lesson plan for an introductory lesson on imaginary numbers and much of the lesson wasn't practical. The lesson went over almost every aspect of imaginary numbers, from solving equations to operations with complex numbers to real-world applications. The lesson was mainly teacher-centered and did not seem very engaging from the point of view of a student. I was trying to see if there were any activities or ideas that I could incorporate into my lesson, but I didn't really find the lesson plan usable. (PST response)

Another PST stated that they did not find ChatGPT's output useful as related to pedagogy, again because the lesson was teacher-centered:

> ChatGPT suggested that I give key ideas to the students. The students should be the ones who should come up with the idea of graphing and then observe why it's useful. We first create a use for graphing by asking which mathematical model can represent all possibilities, and then have

> the students come up with the idea and start graphing. ChatGPT takes away from this and suggests we should just tell the students instead of having them struggle for it. The ChatGPT lesson also has the students graph the linear equation before asking for decimals and fractions in the guesses. Graphing the linear equation would then show the students that fractions and decimals can be used and would take a major part away from the discussion. ChatGPT suggests that the students should be given and shown important parts of the lesson, which prevents the productive struggle. (PST response)

A PST's comment related to content stated:

> When I asked ChatGPT for some real-life examples of experimental probabilities, it gave some examples that are not representative of experimental probabilities. [Tossing a paper] cup can be an example of experimental probability because it's not a "fair" objective that all outcomes have equal chances to occur, so it doesn't have a theoretical probability. But examples provided by ChatGPT such as coin toss, dice roll, and spinner game are some typical examples of theoretical probability because they have "fairness" and their outcomes usually have equal chances to occur. Thus, ChatGPT may be useful for building an argument, but it is not very useful for mathematical concepts. (PST response)

The following comment captured what several PSTs mentioned about what was not useful about ChatGPT's output, including issues with aspects of the plan (mentioned by 14 PSTs) and being teacher-centered (mentioned by 3 PSTs).

> ChatGPT is not designed to provide a structured curriculum or sequence of lessons. While ChatGPT can answer questions and provide explanations, it doesn't have the interactive features and engagement that a dedicated math education platform or a teacher can offer. As you can see, the lesson plan ChatGPT provides is a very teacher center lesson (sic). And, ChatGPT doesn't have to think about the type of student it is interacting with, it doesn't consider motivation. (PST response)

In conclusion, the PSTs who were the subjects of the study found ChatGPT useful in a limited way, and often incorporated aspects of ChatGPT's output into the final version of their mini-lesson, but never used the output as generated by the chatbot. The PSTs were rather adept in identifying pedagogical issues with the output, especially noting when the suggested lesson was teacher-centered. However, although the ChatGPT mathematical output was incorrect in 13 of 19 coded instances, only one of the PSTs accurately identified the incorrect mathematical output, even when they solved the problem correctly themselves, the rest attributing the incorrect solution from ChatGPT as simply being a different approach to the problem. Discussion and limitations of these results follow.

Discussion

The results from this study revealed that PSTs engaged in reflective practices when interpreting and using output from ChatGPT for both pedagogy and mathematics content, but were more accurate in their assessments of output related to pedagogy than they were of mathematical output. These results also revealed important limitations in ChatGPT's usefulness in correctly solving mathematics problems, and for aspects of lesson planning. The emerging literature related to chatbots in educational contexts emphasizes the importance of critique and evaluation of ChatGPT output (Knoth et al., 2024; Trust et al., 2023, van den Berg & du Plessis, 2023; Yilmaz & Yilmaz, 2023), which can be related to the reflective process. In this section, we will first discuss how PSTs reacted to and reflected about ChatGPT's lesson planning suggestions, then the PSTs' reflections and reactions related to ChatGPT's mathematical output, followed by recommendations for the field and limitations.

The Artzt framework, which was familiar to the PSTs in the study, emphasizes the need for critical reflection across all phases of teaching, including planning, monitoring, and evaluating, and Celik et al. (2022) found that AI can be useful for supporting teachers in each of these phases. Knoth et al. (2024) state the importance of users of AI to have "the ability, knowledge, and skills to understand, monitor, effectively interact with, and critically reflect on AI-based technologies" (p. 3), while also noting that it is "essential for teachers to acquire AI-related skills as part of their digital teaching skills" (p. 8). The results of this study support these findings.

The PSTs in this study were able to be reflective and critical about ChatGPT's output as related to pedagogy. In fact, there was evidence that the PSTs analyzed the pedagogical output through the lens of teaching the overarching mathematical tasks of the lesson, most frequently discussing the sequencing and difficulty levels in the lesson, and the motivational strategies (Artzt, 1999). The PSTs found output such as structured outlines, suggestions for each section of the lesson, prior knowledge, possible mathematical misconceptions, and real-life applications suggested by ChatGPT to be helpful during the preactive stages of the lesson. They found generated homework assignments and assessments to be repetitive, and that teaching approaches were most often teacher-centered. They suggested that ChatGPT's responses revealed its lack of knowledge of the students and how they learn. This demonstrated the PSTs' own beliefs about teaching and the importance of knowledge of students in lesson planning, which may have been influenced by the ongoing use of the Artzt framework.

Given the well-documented difficulties that PSTs have in lesson planning (Cevikbas et al., 2024; Lilly et al., 2024; Lim et al., 2018; Sahin-Taskin, 2017), ChatGPT appears to be capable of assisting PSTs in planning in a limited way. Although the PSTs found many issues with ChatGPT's suggestions for their lessons, analysis also revealed that 20 of the 22 PSTs in this study made changes to their lessons based on ChatGPT's output. They found suggestions for definitions, objectives, prior knowledge, student misconceptions, and grammar to be useful. They also noted many aspects of the output to not be helpful, again revealing the importance of reflection when interacting with ChatGPT.

When interacting with the mathematical output, results indicated that of the 19 coded solutions for PSTs and the associated ChatGPT output, 15 of the PSTs solved their task correctly while ChatGPT solved it correctly only six times. While it would not necessarily be expected that a PST who solved a problem incorrectly themselves would identify the incorrect response of ChatGPT, it was somewhat surprising that of the 10 PSTs who solved their problem correctly while ChatGPT did not, only one of the PSTs identified ChatGPT's solution as incorrect. Although it is important to mention that the PSTs were not specifically asked to evaluate the correctness of ChatGPT's output, they were asked to compare the solutions, and it is not unreasonable to expect that they would be able to identify an incorrect response rather than attribute the "different" solution as simply an alternate approach.

This raises the question of whether the same PSTs would have had the same attribution if the same mistake or misconception had been proposed by one of their students, or even a peer. Aharoni et al. (2024) may provide some insight given their findings that people found moral guidance generated by ChatGPT to be superior to that of human moral reasoning, cautioning that, "people may uncritically accept potentially harmful moral guidance from AI" (p. 1). These findings may extend to people's perception of other types of output, including mathematics, although this is weakened by the fact that the PSTs *were* able to be critical about pedagogy.

The authors have hypothesized that the PSTs' lack of critical evaluation of mathematics may be related to the non-traditional tasks and their use in an unfamiliar way (i.e., as launch problems for lessons). However, it is more likely that, while they already had a framework for analyzing pedagogy and teaching, they did not have an explicit framework for critical analysis of the mathematics solutions. While it may be impossible to determine, this finding may reveal PSTs' beliefs about their own content knowledge. That is, they were more trusting of ChatGPT's mathematical output because they were not certain about their own solutions. Their lack of criticism of ChatGPT could be attributed to their lack of depth of knowledge of the mathematics content, although this would require further research and study to ascertain.

Recommendations and Conclusions

The results of this study indicate that GAI such as ChatGPT has a place in mathematics teacher preparation as long as PSTs insightfully and critically reflect on the output they receive. Colleges, universities, and teacher preparation programs might consider that GAI can be useful, but not as a self-standing solution. What we have noted in this study is that the accuracy of PSTs' evaluation of the output may be related to their own beliefs and knowledge, which appeared to be stronger as related to pedagogy than to mathematics, which may be because they had already been using the Artzt (1999) framework to analyze their own work. Their beliefs about the importance of student-centered lessons were revealed through their reflections about the pedagogical output, which is likely related to their use of the Artzt (1999) framework.

As related to the mathematical tasks, PSTs' beliefs in their own abilities to evaluate these mathematics tasks may have played a role in how they critiqued the output produced by ChatGPT. Their reflections revealed that they did not always identify incorrect mathematical output, which led them to accept incorrect answers and faulty mathematical language produced by ChatGPT as "different approaches". It seemed, however, when it came to pedagogical approaches their beliefs about and their own understanding of good pedagogy allowed them to be more critical and accurate in their reflections. This may also be attributed to their ongoing use of a framework for recommended pedagogy. Therefore, another framework for evaluating mathematical solutions might be helpful when using GAI.

Limitations

There were several limitations to this study. We were not able to recruit widely for this study, and it was necessary to examine a convenience sample with a small sample size. Results may be different for other groups of PSTs. We were not able to monitor how the PSTs used ChatGPT during the times when they were encouraged but not required to use the chatbot. Further, the participants worked on a variety of math problems, and the outcomes might have been different had they all worked on the same problem and taught the same lesson. Finally, the fact that these problems were mostly non-routine could have skewed the PSTs' beliefs in their ability to correctly solve these problems. PSTs might have been better able to recognize errors in ChatGPT's output if the problems were routine, and might have been more inclined to do so if they were explicitly asked to check for correctness.

Author Contribution The study conception and majority of the literature review, writing, data analysis and conclusions were completed by the first author. The second and third authors proofread and edited the paper and contributed to data analysis. All authors contributed to the data collection and instrument design.

Data Availability The data are unable to be shared publicly because they are connected to specific identifiable accounts, and sharing this could compromise the privacy of the participants.

Declarations

Ethics Approval and Consent to Participate On 11/01/2023, it was determined that research protocol 2023–0690-QC meets the criteria for exemption, in accordance with CUNY HRPP Procedures.

Competing Interests The authors declare no competing interests.

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